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Address standardization using the natural language process for improving geocoding results

Dilek Küçük Matci*, Uğur Avdan

Research Institute of Earth and Space Sciences, Anadolu University, Iki Eylul Capmus, 26470 Eskisehir, Turkey

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ABSTRACT

Geocoding is a tool that can be used in many areas such as the development of disaster prevention systems, crime mapping and the monitoring of communicable diseases, and which has gradually gained importance. However, the use of geocoding is not yet possible in some areas where it could serve as an effective tool, for various reasons such as inconsistencies in address formats, including inaccurate numbering systems, misspellings, the use of abbreviations and a lack of data that refers to the geocoding process. This study seeks to address these problems by way of a standardization process. To that end, it employs a method that decomposes addresses used as input data in geocoding, identifies spelling mistakes and abbreviations, and reorganizes the addresses through the Natural Language Process (NLP). As test data, the addresses of primary schools in the district of Eskisehir are taken. First the geocoding process is performed on the data set, using both Google geocoding API and ArcGIS geocoding API. Then, the addresses are reformatted into three address formats by applying standardization processes. Geocoding is performed on the re-formatted addresses and the results compared to the non-standardized results. The standardization used is shown to make a significant improvement in the accuracy of the geocoding results. The method used in this study is significant not only in increasing the accuracy of the geocoding process, but also in sustaining its wider use.

1. Introduction

An address is information that allows a specific location to be reached by predetermined directives. It is essential that addresses are converted into coordinates when analyzing incidents such as epidemics, crimes, or accidents (Davis & Fonseca, 2007). The conversion of text-based postal addresses to geographic coordinates is known as geocoding.

Address geocoding is a tool which can be used to digitally locate places on a map based only on addresses. It allows us to locate addresses downloaded from various computer systems on various types of digital maps in various situations, such as the investigation of epidemics (Tassinari et al., 2008; Wey, Griesse, Kightlinger, & Wimberly, 2009), crime mapping or analysis systems (Ratcliffe, 2004; Scribner, Cohen, Kaplan, & Allen, 1999), disaster and risk management systems (Johnson, Stanforth, Lulla, & Luber, 2012), and political science (Haspel & Knotts, 2005). It, however, depends on the accuracy and format of the addresses.

As the accuracy of the geocoding process depends on the format of the addresses, it is of vital importance to have not only correct and precise input addresses, but also to have them written in a certain standard way. The metrics commonly employed in evaluating the quality of geocoding results are completeness, positional accuracy, and repeatability (Zandbergen, 2008).

The accuracy of the geocoding process can be influenced by various factors, such as the geographical areas of the addresses, the quality of the reference database, match scores and geocoding algorithms. Previous research has shown that geocoding results are more accurate in urban areas than rural areas ((Bonner et al., 2003), (Cayo & Talbot, 2003),(Ward et al., 2005)), and geocoding results are more accurate for community addresses than commercial addresses (Zandbergen, 2008).

In the case of Turkey, the biggest challenge in achieving accurate geocoding results is inaccuracies in the input address data. Errors in enumeration work, the use of too many address components, and the lack of a specific address standard can all lead to confusion (Cbsgm, C. B. S. G. M., 2009). Research examining address data in Turkey and performing geocoding, has indicated that mismatched addresses result from errors in door numbers, the incompatibility of street names, incomplete addresses, misspellings, incomplete offset data, typographical errors, and improper formats (35, 28, 15, 9, 5, 4 and 4%, respectively) (Yildirim, Yomralioglu, Nisanci, & Inan, 2014).

The scale of the problem becomes evident when one compares the

* Corresponding author.
E-mail addresses: dkmatci@anadolu.edu.tr (D. Küçük Matci), uavdan@anadolu.edu.tr (U. Avdan).

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Table 1
Geocoding results obtained from the data.

	Geocoding results obtained from addresses of schools in New Jersey(%)	Geocoding results obtained from addresses of schools in Eskishehir (%)
≤100	82.1	46.7
100–200	9.4	19.3
200–400	3.3	14.7
400–600	1.2	6.0
600–800	0.0	2.7
≥800	4.1	10.7

geocoding outcomes based on school addresses in New Jersey, US, with those of school addresses in Eskisehir (Eskişehir), Turkey, which is the area used as the case study in this research. Geocoding has been performed by using 264 schools addresses in New Jersey and 233 schools addresses in Eskishehir, all provided by the official web pages of the respective Ministries of Education (Milli Eğitim Bakanlığı, 2016; Schools, 2016). The results of the calculation of the distances between the obtained coordinates and the actual coordinates of the schools is shown in Table 1.

When the geocoding process is applied to the 264 school addresses in New Jersey, 246 (93.2% of all addresses) coordinates can be obtained. 82.1% of these coordinates are located at a distance less than 100 m from the actual location. On the other hand, when the geocoding process is performed on the 233 school addresses in Eskishehir, only 150 coordinates can be obtained (65% of coordinates). 46.7% of these coordinates are located at a distance less than 100 m from the actual location. According to these results, the success rate of geocoding performed with US addresses is 82.1%, while this ratio is 46.7% for the addresses in Turkey. Considering the fact that address standardization is part of the address matching process, the latter success rate is extremely low. For this reason, a customized standardization process based on specific regions is a necessity.

The process of translating manually written addresses into a certain digital format is known as address standardization. Unlike human written addresses, in the standardized format every section of the address, e.g. road, city, etc., can be separately identified (Abbasi, 2005). This study seeks to increase the success rate by the standardization of addresses through an innovative use of Natural Language Processing, which uses artificial intelligence methods to communicate with the computer in natural language.

Chomsky's study of Natural Language Processing (Chomsky, 1986) is highly influential, and Natural Language Processing has been used in a number of pieces of research for various purposes, including correcting misspellings (Kukich, 1992), summarizing text (Hu & Liu, 2004), voice interaction with a computer (Weber, 2003), and translating between natural languages (Nitta, Okajima, & Yamano, 1987).

Improving geocoding process results is not an unknown topic. Research which examines concerns over accuracy (Dickson et al., 2017), has undertaken geocoding with three tools and reached match rate results between 82% and 88%. This research suggests that parsing/standardizing tools can greatly improve the results of geocoding and seeks to improve the results by using different tools in different parts of the geocoding process. For example, while one tool is used to determine the input addresses, another is used to geocode addresses at street level. The same strategy is used in another study (Yang, Bilaver, Hayes, & Goerge, 2004) to test three geocoding tools and demonstrate that each tool provides better results in different parts of the geocoding process. As a result, the research concludes that the best strategy would be to combine the three methods.

In another study (Tian et al., 2016), an optimized address matching method for Chinese geocoding with three components, address modeling, address standardization and address matching, is proposed. The suggested model is structured around a standardization process based on an address tree model proposed in a previous study (Mengjun, Qingyun, & Mingjun, 2015). In this model the input address string is parsed and organized as a collection of address elements X and a semantic collection S. The root node is created and address element X1 is extracted. Finally, all the semantic elements in S1 associated with X1 are traversed to create address semantic nodes and connected to the root node. It is concluded that 60.4% of the addresses are matched accurately for company data, 86% of the addresses are matched successfully at a matching degree > 60%, and the corresponding matching rates for disease data are 49.3% and 98.6%, respectively, at the same matching degrees. These results are obtained by using the geocoding service created by them and the local data in their study area. It is pointed out that the reasons for there being different results for different data, include the company data being prepared in a more regular manner, and many of the addresses in the disease data not being located in their study area.

Advanced probabilistic methods such as Hidden Markov Models (HMM) are used to deal with misspellings and misplacements in some studies (Peter Christen, Churches, & Willmore, 2004; Christen, Willmore, & Churches, 2006; Churches, Christen, Lim, & Zhu, 2002). (Christen et al., 2004) propose a technique based on HMM to clean and standardize addresses and reach a 72.87% address level matching using Geocoded National Address File (GNAF) data.

Our study does not perform a simple standardization process. The model developed in this research enables the correction of spelling mistakes in input data, completion of the missing data in the addresses, and most importantly, by examining which address structure gives the higher success rate in the geocoding process, it offers a new address format. In order to improve the results of the geocoding process, the test dataset is standardized in three formats (PTT, Google, ArcGIS) using an NLP-based approach. For this purpose, parser, semantic analyzer and generator modules contained in the NLP systems are established. With these modules, raw address texts are parsed into address components; misspellings, abbreviations and incorrect address formats are corrected; and address data is created in the new format. In order to examine the impact of the address formats on the results of the geocoding process, the geocoding process is applied to the reproduced address data in the various formats and the raw address data. The online world geocoding services of google geocoding API and ArcGIS geocoding API are used to perform the geocoding operations.

2. Materials and methods

2.1. Dataset

In the study, the addresses of 233 primary schools located in the Eskishehir (Eskişehir) region are selected as a test dataset. These addresses are obtained from the official websites of the schools. Examples of the data are given in Table 2.

Addresses in Turkey include words that indicate areas, such as street (sokak), neighborhood (mahalle), and avenue (cadde or bulvar). However, there is neither a single address format for the input address nor a specific standard for the abbreviations. For example, while one uses the abbreviation “Sk.” to denote the word sokak, the other uses “Sok”. More importantly, some of the addresses omit the information regarding the province. In the study, the actual coordinates of the schools are obtained using Google Earth on top of the school buildings, to check the accuracy of the coordinates found through the geocoding process.

This study is conducted based on school addresses because in Turkey schools are also used for purposes other than education, for instance, in the nationwide central examinations run by the Ministry of Education. In many of these examinations, candidates are assigned to a school buildings located in provinces other than where they live, which poses a problem in finding the location of the examination centre. With the method proposed in this study the addresses of schools, hotels or

Table 2
Test data.

School name	Address
Çamlıca Ticaret Odası İlkokulu	Çamlıca Mahallesi Cihangül Sokak No4 TEPEBAŞI/ESKİŞEHİR
Çamlıca Ticaret Odası İmam Hatip Ortaokulu	Çamlıca Mahallesi Okumuşlar Sokak No1 Tepebaşı Eskishehir
Çukurhisar İlkokulu	Yeni mah.8. sokak No1 Merkez Çukurhisar Tepebaşı/ESKİŞEHİR
Çukurhisar Ortaokulu	Yeni Mah. Fatih Cad.No12A Çukurhisar Tepebaşı/ESKİŞEHİR
Öğretmen Esra Akkaya Anaokulu	Yenikent Mah. Küme 18
Ülkü İlkokulu	Yenibağlar Mahallesi Özbek Sokak No1 Tepebaşı ESKİŞEHİR
İstiklal Ortaokulu	Uluönder Mahallesi Aysev Sokak No1 Tepebaşı/ESKİŞEHİR
İbrahim Karaoğlanoğlu İlkokulu	Kurtuluş Mah. Aydınlık Sok. No10 26090 Eskishehir/TÜRKİYE
İbrahim Kurşungöz Rehberlik ve Araştırma Merkezi	Akarbaşı M. Ada S. 35/A Odunpazarı ESKİŞEHİR
İki Eylül Ortaokulu	Cumhuriye Mh. Temel Sk. No5 Tepebaşı/ESKİŞEHİR

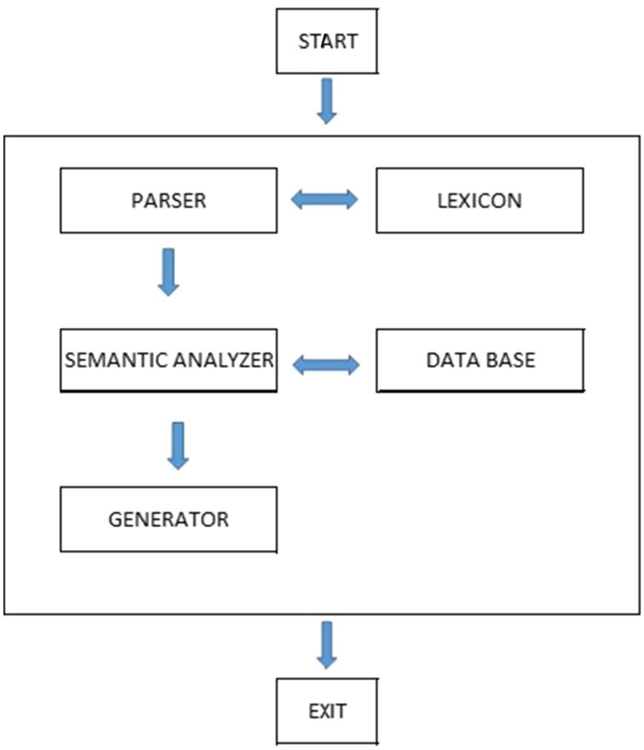

Fig. 1. Block diagram of general NLP systems.

Dictionary 1	Dictionary 2
caddesi	ADA
CAD	ADAÇAL
cad.	ADAK
cd.	ADAKALE
c.	ADAKENT
mevkisi	BÜYÜKDERE
mev.	BÜYÜKKAYA
mevkii	BÜYÜKLER
mevki	CAFER
mahallesi	CAHİDE
mahlesi	CAHİT KÜLEBİ
mahalle	CAMCI
Mahlesi	CAMCILAR
MAH	CAMİ
mah.	CAMİ ÇIKMAZI
Mhl.	CAMİALTI
mh.	LÜLECİ
mh	LÜLECİLER
m.	LÜLETASI
sokaği	LÜLETAŞI
sokağı	LÜTFİYE
sokagi	M. K. ATATÜRK
sokak	MAARİF
sokak.	MADEN
Sok.	MADENLER
sk	MAĞRUR
sk.	MAHİR
skk	MAHMUT AKAY
skk.	MAHMUT PEHLİVAN
s.	MAHMUT SEVKET ESENDAL

Fig. 2. Samples from the dictionaries.

hospitals could be standardized to achieve a higher success rate in the geocoding process.

2.2. Methods

2.2.1. Natural language processing

Natural language processing (NLP) is a science that uses artificial intelligence methods to communicate with the computer in natural language. Information extraction, automatic indexing, and summarizing or generating text in desired formats are research topics in NLP. Broadly, natural language text processing takes partial information from a large textual structure or receives information to be used for a specific purpose. Automatic text processing systems usually get text in a specific format and convert the output to another format (Liddy, 1998). Dictionaries are used to determine the characteristics of words, detect language, detect words and expressions and separate cues using lexical and syntactic processes (Chowdhury, 2003).

There are five basic elements of natural language processing systems in general, the parser, dictionary (lexicon), semantic analyzer, knowledge base and generator. Fig. 1 shows the interaction of these five elements. The parser module creates a parsing tree by analyzing the sentence, and the lexicon is a structure which contains all the words that have to be recognized by the program. The semantic analyzer module tries to identify what the sentence means given the knowledge base, prepare a response according to the sentence given, and load it into memory. The interface shows the proper response to the user. This response could be the result of the query in database management. The main generator systems used in NLP show certain patterns, stored for certain words and phrases, to the user (Elektrik-Elektronik, 2015).

It is possible to encounter problems resulting from incorrect spelling in the address matching process implemented by the semantic analyzer. In order to avoid these problems, matching algorithms such as the Levenshtein Distance Algorithm and the Match Rating Compute Approach are used in this study. These algorithms are described in the following sections.

2.2.2. Levenshtein distance algorithm

The Levenshtein Distance Algorithm, developed by Levenshtein (Levenshtein, 1965), is the number of basic operations needed to convert one string to another (Schimke, Vielhauer, & Dittmann, 2004).

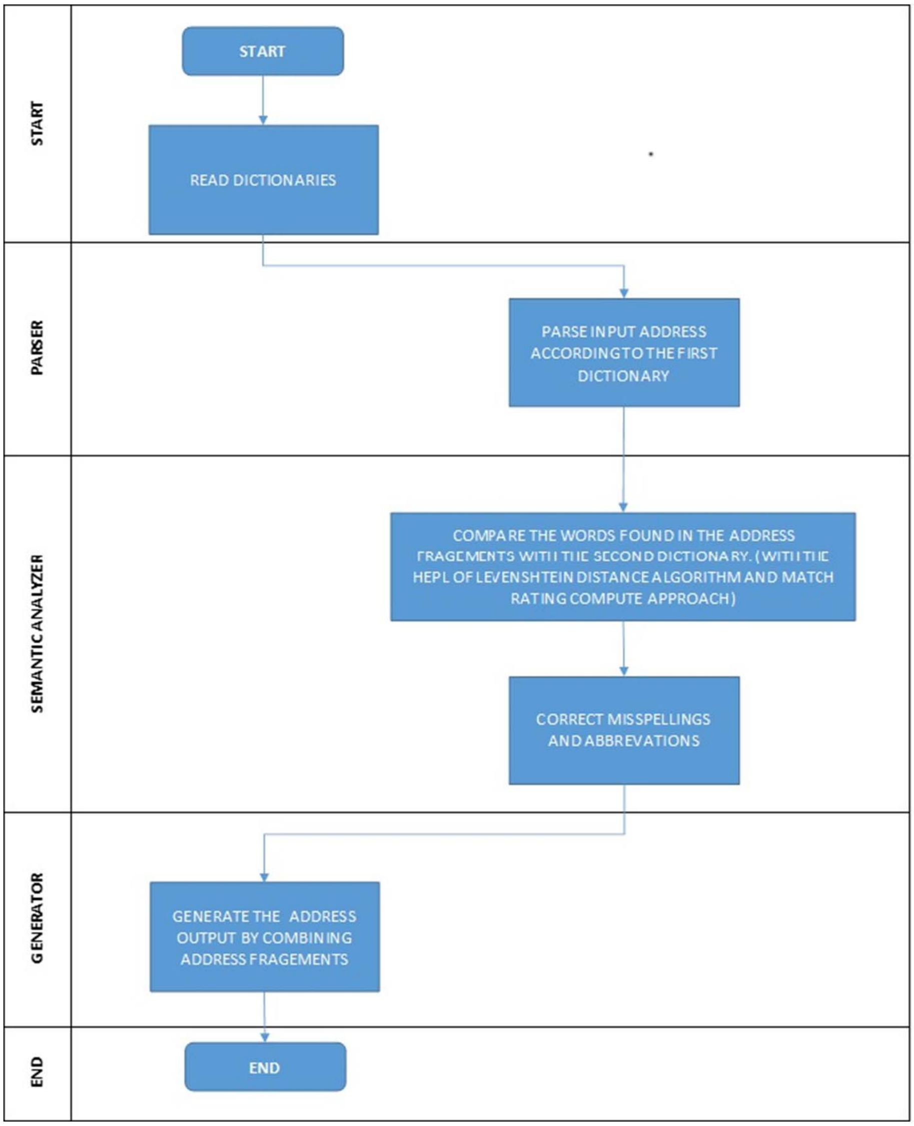

Fig. 3. Standardization process.

These operations consist of inserting or removing a letter, inserting one letter instead of another or changing the order of two letters side by side. Various combinations of these four operations convert one text to another. The solution using the minimum number of operations gives the distance between two texts (Bard, 2007).

Levenshtein distance (LD) is defined by Eq. 1 for texts A and B:

$$\mathrm{LD}\,(\mathrm{A},\mathrm{B}) = min\{\mathrm{a(i)} + \mathrm{b(i)} + \mathrm{c(i)}\}. \tag{1}$$

In Eq. 1, B represents the text that is obtained from A, a(i) represents replacements, b(i) represents insertions, and c(i) represents deletions of individual symbols (Kohonen & Somervuo, 1998). The Levenshtein Distance Algorithm is generally used to find errors and deficiencies in text, in programs that check spelling rules, search engines and linguistics.

2.2.3. *Match rating compute approach*

The second method used in this study is the match rating compute approach. This method is a phonetic algorithm developed by Western Airlines in 1977 to compare and index homonymous words (Moore, 1977). In this method, the vowels are deleted from the compared words, then the second consonants are deleted. The first three and last three characters are combined and words are reduced to a maximum of six characters. After this preparation, the following comparison rules are applied:

1. There is no similarity, if the difference in length is more than three between the two words that have been created.
2. The minimum rating must be determined by calculating the sum of the lengths of the words that have been created.
3. The same characters should be deleted proceeding from left to right on both words.
4. The same characters should be deleted proceeding from right to left on both words.
5. The degree of similarity is obtained by subtracting the remaining number of characters from 6.

2.2.4. *Applied method*

In this study, NLP-based methods are used to find the address components of the data, parse, edit and bring them together. In this context, an application is developed in C# language. In the developed application, parser, semantic analyzer and generator modules are formed. Two dictionaries are created by the authors to be used in the parsing and semantic analysis processes. The first dictionary consists of the words that are likely to be used to define all the address components found in the address declarations. For example, for the word “mahalle”, that is used to define the neighborhood in Turkish, similar words and abbreviations, such as “mahallesi”, “mah”, “mh.”, and “m.”, are added to the dictionary. Address declarations are examined, and address components within the addresses are identified, using this dictionary.

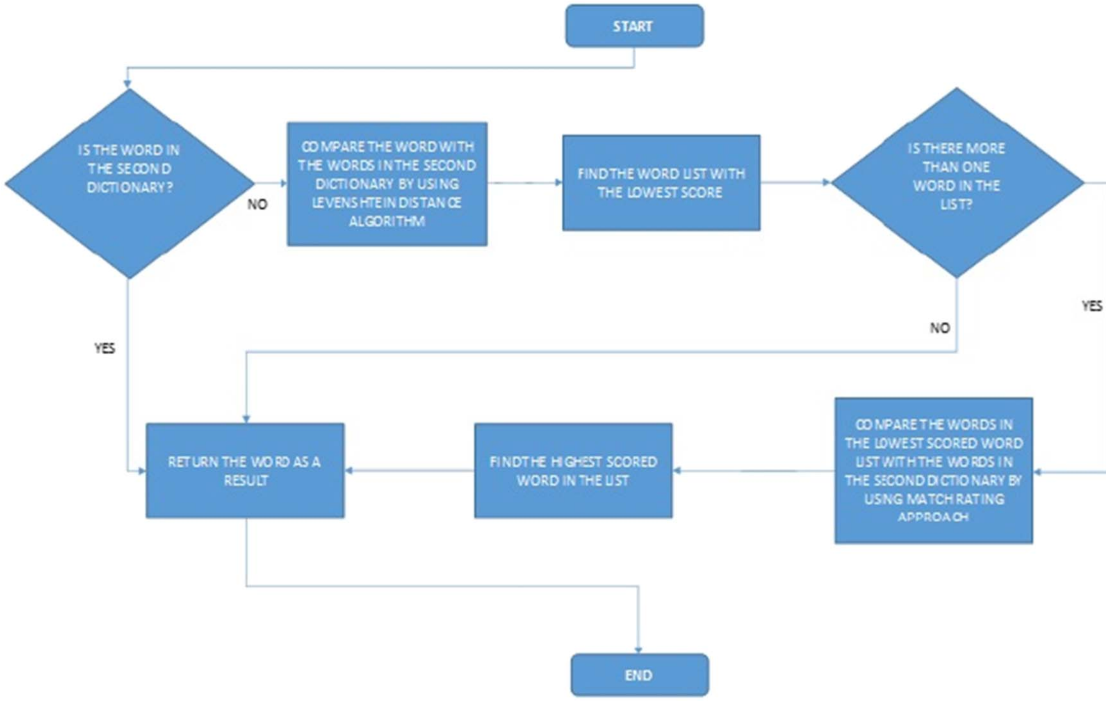

Fig. 4. Process of the semantic analyzer module.

STAGE	THE STATE OF THE ADDRESS	THE STATE OF THE ADDRESS	ADDRESS FORMAT
Start	HOŞNDYE MAHLESİ.SAĞIN SOKAK.NO32	Address Line	
Parser	“HOŞNDYE MAHLESİ”	Address Fragments	-
	“SAĞIN SOKAK”		
	“NO”		
	32		
Semantic analyzer	“HOŞNUDİYE MAHALLESİ”	Corrected Address Fragments	-
	“SAĞIN SOKAK”		
	“NO”		
	32		
Generator	HOŞNUDİYE MAH. SAĞIN SOK. NO: 32 TEPEBAŞI ESKİŞEHİR, TÜRKİYE	True Address Line	Ptt format
	HOŞNUDİYE, SAĞIN SK. NO: 32, TEPEBAŞI/ ESKİSEHİR, TURKEY		Google format
	HOŞNUDİYE, SAĞIN SOKAK 32, TEPEBAŞI, ESKİSEHİR		Arcgis format

Fig. 5. States of the address line.

Table 3
Addresses standardized according to the PTT format.

Input address	Output address
Esentepe Mah. Akçağ Sok. No3 Tepebaşı/Eskishehir	Esentepe Mah. Akçağ Sok. No:3 Tepebaşı Eskisehir,Türkiye
Büyükdere Mh. Nisan Sk. No32 Odunpazarı/Eskishehir	Büyükdere Mah. Nisan Sok. No:32 Odunpazarı Eskisehir,Türkiye
Emek Mh. Çetinsoy Sok.No51 Odunpazarı/Eskishehir	Emek Mah. Çetinsoy Sok. No:51 Odunpazarı Eskisehir,Türkiye
Çamlıca Mahallesi Yetim Sokak No 4 Tepebaşı/Eskishehir	Çamlıca Mah. Yetim Sok. No:4 Tepebaşı Eskisehir,Türkiye
Arifiye Mahallesi Tayfun Sokak No10 Odunpazari/Eskishehir	Arifiye Mah. Tayfun Sok. No:10 Odunpazarı Eskisehir,Türkiye
Sütlüce Mh. Uluzafer Cd. No87 Tepebaşı/Eskishehir	Sütlüce Mah. Uluzafer Cad. No:87 Tepebaşı Eskisehir,Türkiye
Ihlamurkent Mh. Oymakbaşı Sok. No5 Odunpazarı Eskishehir	Ihlamurkent Mah. Oymakbaşı Sok. No:5 Odunpazarı Eskisehir,Türkiye
Şirintepe Mahallesi Metal Sokak No.2 Tepebaşı/Eskishehir	Şirintepe Mah. Metal Sok. No:2 Tepebaşı Eskisehir,Türkiye
Vişnelik Mah. Ş.Yusuf Turgut Sok. No1 Odunpazarı/Eskishehir	Vişnelik Mah. Şehit Yusuf Turgut Sok. No:1 Odunpazarı Eskisehir,Türkiye
Gökmeydan Mahhallesi Kağıthane Sokak No51 Odunpazari Eskishehir	Gökmeydan Mah. Kağıthane Sok. No:51 Odunpazarı Eskisehir,Türkiye

The second dictionary consists of the district, neighborhood and street lists of Eskishehir. The street, neighborhood etc. names used in the address data are controlled using this dictionary and spelling mistakes are rectified. Samples from the dictionaries are given in Fig. 2.

The address standardization process consists of three modules. The modules created and the flow of the standardization process are given in Fig. 3.

In the parsing step, words that are in the first dictionary are searched for in the text. The address text is parsed through available words and thereby address fragments are obtained. For example, an address line in the form “HOŞNDYE MAHLESİ.SAĞIN SOKAK.NO32” is divided into address fragments “HOŞNDYE MAHLESİ”, “SAĞIN SOKAK”, “NO”, “32”. Thus, address fragments are created for all address components. These address components represent parts of the address structure, such

Table 4
The geocoding results obtained from the raw data and standardized data according to the PTT.

Address format	PTT			
Technology	Google		Arcgis	
	Non-standardized address(%)	Standardized address(%)	Non-standardized address(%)	Standardized address(%)
≤100	46.7	52.8	35.8	48.5
100–200	19.3	12.6	10.6	15.3
200–400	14.7	7.6	14.6	14.9
400–600	6.0	1.7	7.5	4.4
600–800	2.7	1.7	6.6	3.5
≥800	10.7	23.8	24.8	13.5
Total acquired	64.4	99.14	97	98.3

Table 6
The geocoding results obtained from the raw data and standardized data according to the Google Geocoding APIs format.

Address format	Google			
Technology	Google		Arcgis	
	Non-standardized address	Standardized address	Non-standardized address	Standardized address
≤100	46.7	52.8	35.8	45.2
100–200	19.3	12.6	10.6	15.4
200–400	14.7	7.8	14.6	15.8
400–600	6.0	1.7	7.5	4.8
600–800	2.7	2.2	6.6	3.5
≥800	10.7	22.9	24.8	15.4
Total acquired	64.4	99.1	97.0	97.9

as neighborhood, street or avenue.

At the understanding stage, mistakes found in the words (neighborhood, street names, etc.) are corrected. This process is performed by comparing the words contained in the address fragments with the words in the second dictionary. The major problem faced at this stage is unsuccessful results of matching the words in the dictionary with the words in the address fragments. This results from the incorrect spelling of words. For example, the semantic analyzer module gives incorrect results because of the use of "Hoşndye" instead of "Hoşnudiye". The Levenshtein distance method is used to prevent errors like this. The neighborhood, street, etc. names and the words in the address fragments are compared, one by one, using the Levenshtein Distance Algorithm. Then, a similarity score is given for all the words found in the dictionary. This score shows the number of operations required to convert one word to another. Finally, the lowest scored word is accepted as the result. The detailed process steps of the semantic analyzer module are given in Fig. 4.

The Levenshtein distance method generally provides accurate results, but in some cases more than one word with an equal lowest score can be found. Therefore, the search has to be repeated with a different algorithm using the generated candidate result list. Hence, the words found in the results list and the words contained in the address fragments are compared with the Match Rating Compute Approach. In the comparison process, a similarity score is given to all candidate words and the highest scoring word is returned as the result.

In the generator stage, output address data is obtained by combining the neighborhood, village, street etc. names in the specific address format. The state of the sample address through the stages is shown in Fig. 5.

When this process is performed, it is important that the address components are ordered in the desired address format. Additional features such as abbreviations, punctuation marks, and spaces found in the address line are arranged according to the desired address format.

After the standardization process, the geocoding step is performed by the application, which uses the default reference data of the ESRI and Google geocoding system. It is known that these technologies use up-to-date reference data in their documents. Address data is taken from the official web site of the Ministry of the Education.

3. Results and discussion

In accordance with the algorithms developed, the application, developed in the C# language, and the input address data used in the study are standardized. Three addresses are generated with the purpose of testing whether the format of the address affects the result of the geocoding process. These formats are Turkish National Post Telephone Telegraph (PTT) address format, Google geocoding API address format and the ESRI ArcGIS API address format. First, the input address data is standardized according to the format produced in accordance with "address writing rules on the envelope" (PTT, 2009) set by the General Directorate of Post, Telegraph and Telephone. Table 3 gives examples of the addresses generated.

The geocoding process is applied to the input address data and the address data created through the standardization process. Google geocoding API and ArcGIS geocoding API are used for the geocoding process. As these two geocoding services use different reference databases, as is expected, they provide different outcomes when applied. However, we do not seek to compare the results offered by these geocoding services. We rather compare the results of geocoding operations obtained before the address standardization process with those obtained after the standardization.

The distance between the geocoded and real coordinates are calculated, and the results are shown in Table 4, which also gives the minimum and maximum distances found by both geocoding tools. The calculated distances are classified according to the specified interval. Generally, if the distance calculated is less than 100 m, it is considered a

Table 5
Addresses standardized according to the Google geocoding API format.

Input address	Output address
Uluönder Mahallesi Nezih Sokak No 3 Tepebaşı Eskishehir	Uluönder, Nezih Sk. No:3, Tepebaşı/Eskisehir,Turkey
Şirintepe Mahallesi Erkanat Sokak No2 Tepebaşı/Eskishehir	Şirintepe, Erkanat Sk. No:2, Tepebaşı/Eskisehir,Turkey
Sütlüce Mhl. İstiklal Çıkmazı No8 Tepebaşı/Eskishehir	Sütlüce, İstiklal Çıkmazı, No:8, Tepebaşı/Eskisehir,Turkey
Gökmeydan Mah. Sarper Cad. No 20 Odunpazarı/Eskishehir	Gökmeydan, Sarper Cd. No:20, Odunpazarı/Eskisehir,Turkey
Şarhöyük Mahallesi Bülbüller Sokak No 56 Tepebaşı Eskishehir	Şarhöyük, Bülbüller Sk. No:56, Tepebaşı/Eskisehir,Turkey
Kozkayı Mah. 165. Sokak. No2 Tepebaşı/Eskishehir	Kozkayı, 165. Sk. No:2, Tepebaşı/Eskisehir,Turkey
Karapınar Mahallesi Terziler Sokak No1 Odunpazarı Eskishehir	Karapınar, Terziler Sk. No:1, Odunpazarı/Eskisehir,Turkey
Şirintepe Mahallesi Örme Sokak No117 Tepebaşı/Eskishehir	Şirintepe, Örme Sk. No:117, Tepebaşı/Eskisehir,Turkey
Uluönder Mh. Yakapınar Sk. No2 Tepebaşı/ESKİŞEHİR	Uluönder, Yakapınar Sk. No:2, Tepebaşı/Eskisehir,Turkey
Erenköy Mahallesi Okullu Sokak No22 Odunpazarı/Eskishehir	Erenköy, Okullu Sk. No:22, Odunpazarı/Eskisehir,Turkey

Table 7
Addresses standardized according to the ArcGIS geocoding API format.

Input address	Output address
Batıkent Mah. Nadire Sk.No3 Tepebaşı	Batıkent, Nadire Sokak 3, Tepebaşı,Eskisehir,
Çamlıca Mahallesi Cihangül Sokak No4 Tepebaşı/Eskishehir	Çamlıca, Cihangül Sokak 4, Tepebaşı,Eskisehir,
Deliklitaş Mahallesi Deliklitaş Caddesi No55 Odunpazarı/Eskishehir	Deliklitaş, Deliklitaş Caddesi 55, Odunpazarı,Eskisehir,
Emek Mah. Ertaş Cad. Tarih Sok.No4	Emek, Ertaç Caddesi, Tarih Sokak 4, Odunpazarı,Eskisehir,
Ertuğrulgazi Mahallesi Türeli Sokak No11/1 Tepebaşı Eskishehir	Ertuğrulgazi, Türeli Sokak 11/1, Tepebaşı,Eskisehir,
Gökmeydan Mah. Sarper Cad. No 20 Odunpazarı/Eskishehir	Gökmeydan, Sarper Caddesi 20, Odunpazarı,Eskisehir,
Kırmızıtoprak Mh. Vişne Sok. No 37	Kırmızıtoprak, Vişne Sokak 37, Odunpazarı,Eskisehir,
Orhangazi Mh. Azizbolel Sok.No 43 Odunpazarı Eskishehir	Orhangazi, Aziz Bolel Sokak 43, Odunpazarı,Eskisehir,
Vişnelik Mah. Ş.Yusuf Turgut Sok. No1 Odunpazarı/Eskishehir	Vişnelik, Şehit Yusuf Turgut Sokak 1, Odunpazarı,Eskisehir,
71 Evler Mh. Hacı Arif Bey Sok. No2	71 Evler, Hacı Arifbey Sokak 2, Odunpazarı,Eskisehir,

Table 8
The geocoding results obtained from the raw data and standardized data according to the to the ArcGIS geocoding APIs format.

Address format	Arcgis			
Technology	Google		Arcgis	
Distance to the real coordinates	Non-standardized address	Standardized address	Non-standardized address	Standardized address
≤100	46.7	52.9	35.8	45.2
100–200	19.3	12.7	10.6	15.4
200–400	14.7	7.7	14.6	15.8
400–600	6.0	1.8	7.5	4.8
600–800	2.7	1.8	6.6	3.5
≥800	10.7	23.1	24.8	15.4
Total acquired	64.4	94.9	97.0	97.9

successful result of the geocoding process. Therefore, an improvement in the number of coordinates obtained in this range is considered a success of the application.

When the geocoding process is applied to the 233 addresses that are not standardized using Google geocoding API, 150 (64%) coordinates are obtained. 46.7% of these coordinates are located at a distance < 100 m from the actual location. The number of coordinates obtained increases to 231 when the geocoding process is applied to the 233 standardized addresses. 52.8% of the coordinates obtained are located at a distance < 100 m from the true location. According to these results, a great improvement is shown in the number of coordinates obtained, and the number of coordinates less than 100 m from the true location.

When the geocoding process is applied to the 233 addresses that are not standardized using ArcGIS geocoding API, 226 coordinates are obtained. 35.8% of these coordinates are located at a distance of < 100 m from the true location. This number increases to 48.7% when the geocoding process is applied to the 233 addresses that are standardized.

The second address format created by the generator module is the format that is identified through Google geocoding API, specifically in the section 'formatted addresses' found as a result of geocoding. Examples of the addresses obtained are shown in Table 5.

The results found using both standardized addresses and unstandardized addresses are given in Table 6.

As shown in Table 6, when the geocoding process is applied to the 233 unstandardized addresses using Google geocoding API, 150 coordinates (64.4%) are identified. 46.7% of these coordinates are located at a distance < 100 m from the true location. The number of coordinates increases to 231 (99.1%) when the geocoding process is applied to the 233 standardized addresses. 52.8% of the coordinates obtained are located at a distance of less than 100 m from the true location.

When the geocoding process is applied to the 233 addresses that are not standardized using ArcGIS geocoding API, 226 coordinates are obtained. 35.8% of these coordinates are located at a distance less than 100 m from the true location. This number increases to 45.2% when the geocoding process is applied to the 233 standardized addresses.

The second address format generated by the generator module is the format that is identified through ArcGIS geocoding API, specifically in the "fully matched address" section, found as a result of geocoding. Examples of the addresses obtained are shown in Table 7.

The results obtained from the geocoding process with non-standardized addresses and the results obtained from the geocoding process with standardized addresses are given in Table 8.

As can be seen in Table 8, when the geocoding process is applied to the 233 unstandardized addresses using Google geocoding API, 150 (64.4%) coordinates are obtained. 46.5% of these coordinates are located at a distance < 100 m from the true location. The number of coordinates obtained increases to 221 (94.9%) when the geocoding process is applied to the 233 addresses that are standardized. 52.9% of the coordinates obtained are located at a distance < 100 m from the true location. When the geocoding process is applied to the 233 addresses that are not standardized using ArcGIS geocoding API, 226 (97%) coordinates are obtained. 35.8% of these coordinates are located at a distance < 100 m from the true location. When 233 standardized addresses are used, this number reaches 45.2%.

Some of the addresses that could not be matched as a result of the geocoding process do not include the address component. For example, "Anadolu University Yunusemre Campus". This school is located on a university campus in Eskishehir. This address cannot be standardized because the prepared NLP module looks for address components in the address line. Therefore, neither before nor after standardization, a match could be made.

Overall, the standardization process leads to significant improvements in the number of coordinates obtained by geocoding and the accuracy of the coordinates. Improvement rates vary depending on the technology used and the format of the addresses generated. In the result of the geocoding process, at most 231 coordinates are identified when 233 addresses are used. This number is reached when the addresses are standardized according to both Google geocoding API's "formatted address" field format and PTT address format, using the Google geocoding API. When the number of coordinates that are located at a distance of < 100 m from the true location are analyzed, the best results are achieved when the addresses are standardized according to both Google geocoding API's "formatted address" field format and PTT address format using the Google geocoding API. This, standardization process improves the rate of coordinates that match from 64.4% to 99.1%.

4. Conclusion

In this study, we present a method to eliminate problems (such as typographical errors, misspellings and incorrect formats) that affect the success rate of the geocoding process. In this method, the address data is parsed using NLP methods. Misspellings, omissions or abbreviations are removed with the Match Rating Compute Approach and Levenshtein Distance Algorithm. As a result, the addresses are rearranged into a specific format.

In order to analyze the effects of the method on the results of the geocoding process, a test data set consisting of addresses of primary schools in Eskishehir, is created. The geocoding process is carried out using this sample set of addresses, both before and after the standardization process is applied.

Geocoding is applied to the addresses using Google geocoding API and ArcGIS geocoding API before and after standardization. The results obtained are analyzed, and the standardization process gives a 50% improvement in coordinate capture using Google geocoding API. More importantly, the standardization process provides an improvement of the rate of coordinates that are matched from 64.4% to 99.1%.

These results show that address standardization affects the results of using default world geocoding services and reference data instead of local reference data. It is obvious that using frequently updated reference data taken from a local source with our standardization model would improve the accuracy of geocoding results. This study seeks only to find how address standardization affects geocoding results, not promote or undermine the geocoding systems used.

Eskishehir region is chosen as the study area. However, the suggested method can be applied in any country or city to improve geocoding results. The high success rate obtained using the proposed method is of great importance, particularly for studies conducted in countries with low geocoding success rates. Data obtained by the geocoding process are used in many GIS projects such as investigation of epidemics, crime tracking systems, post-disaster studies and customer tracking systems. Errors in data obtained through the geocoding process cause critical problems in analysis. Therefore, the proposed method is important both in terms of increasing the accuracy of the results obtained by the geocoding process and spreading the use of the geocoding process.

References

Abbasi, R. A. (2005). Information extraction techniques for postal address standardization. *Proceedings of the INMIC 2005: 9th International Multitopic Conference — Proceedings* (pp. 309–314). .

Bakanlığı, T. C. M. E. (2016). *Milli Eğitim Bakanlığına Bağlı Okullar ve Diğer Kurumlar*.

Bard, G. V. (2007). Spelling-error tolerant, order-independent pass-phrases via the damerau-levenshtein string-edit distance metric. *Proceedings of the fifth Australasian symposium on ACSW frontiers — Volume 68 pp. 117-124*. Ballarat, Australia: Australian Computer Society, Inc.

Bonner, M. R., Han, D., Nie, J., Rogerson, P., Vena, J. E., & Freudenheim, A. L. (2003). Positional accuracy of geocoded addresses in epidemiologic research. *Epidemiology, 14*, 408–412.

Cayo, M. R., & Talbot, T. O. (2003). Positional error in automated geocoding of residential addresses. *International Journal of Health Geographics, 2*, 1.

Cbsgm, C. B. S. G. M (2009). *Türkiye Ulusal Coğrafi Bilgi Sistemleri Standartları Belirlenmesi Projesi: Tucbs Metaveri İlke Ve Esaslarının*.

Chomsky, N. (1986). *Knowledge of language: Its nature, origin, and use.* Praeger.

Chowdhury, G. G. (2003). *Natural language processing. Annual review of information Science and technology. 37*, 51–89.

Christen, P., Churches, T., & Willmore, A. (2004). A probabilistic geocoding system based on a national address file. *Proceedings of the 3rd Australasian data mining conference, Cairns*.

Christen, P., Willmore, A., & Churches, T. (2006). A probabilistic geocoding system utilising a parcel based address file. *Data Mining: Theory, Methodology, Techniques, and Applications, 3755*, 130–145.

Churches, T., Christen, P., Lim, K., & Zhu, J. X. (2002). Preparation of name and address data for record linkage using hidden Markov models. *BMC Medical Informatics and Decision Making, 2*(9).

Davis, C. A., & Fonseca, F. T. (2007). Assessing the certainty of locations produced by an address geocoding system. *GeoInformatica, 11*, 103–129.

Dickson, M. M., Giuliani, D., Espa, G., Bee, M., Taufer, E., & Santi, F. (2017). Design-based estimation in environmental surveys with positional errors. *Environmental and Ecological Statistics,* 1–15.

Elektrik-Elektronik, T. (2015). *Doğal Dil İşleme*.

Haspel, M., & Knotts, H. G. (2005). Location, location, location: Precinct placement and the costs of voting. *Journal of Politics, 67*, 560–573.

Hu, M., & Liu, B. (2004). Mining and summarizing customer reviews. *Proceedings of the tenth ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 168–177). ACM.

Johnson, D. P., Stanforth, A., Lulla, V., & Luber, G. (2012). Developing an applied extreme heat vulnerability index utilizing socioeconomic and environmental data. *Applied Geography, 35*, 23–31.

Kohonen, T., & Somervuo, P. (1998). Self-organizing maps of symbol strings. *Neurocomputing, 21*, 19–30.

Kukich, K. (1992). Techniques for automatically correcting words in text. *ACM Computing Surveys (CSUR), 24*, 377–439.

Levenshtein, V. I. (1965). Binary codes capable of correcting deletions, insertions, and reversals. *Soviet Physics – Doklady, 10*, 707–710.

Liddy, E. D. (1998). Enhanced text retrieval using natural language processing. *Bulletin of the American Society for Information Science and Technology, 24*, 14–16.

Mengjun, K., Qingyun, D., & Mingjun, W. (2015). A new method of Chinese address extraction based on address tree model. *Acta Geod. Cartogr. Sin, 44*, 99–107.

Moore, G. B. (1977). Accessing individual records from personal data files using non-unique identifiers. *Final report. Computer science & technology series*.

Nitta, Y., Okajima, A., & Yamano, F. (1987). *Method for automatic translation between natural languages.* (Google Patents).

PTT (2009). *Adres Yazım Kuralları*.

Ratcliffe, J. H. (2004). Geocoding crime and a first estimate of a minimum acceptable hit rate. *International Journal of Geographical Information Science, 18*, 61–72.

Schimke, S., Vielhauer, C., & Dittmann, J. (2004). Using adapted levenshtein distance for on-line signature authentication. *Pattern recognition, 2004. ICPR 2004. Proceedings of the 17th international conference on pp. 931-934. IEEE*.

Schools, P. P. (2016). *Paterson public schools-school list*.

Scribner, R., Cohen, D., Kaplan, S., & Allen, S. H. (1999). Alcohol availability and homicide in New Orleans: Conceptual considerations for small area analysis of the effect of alcohol outlet density. *Journal of Studies on Alcohol, 60*, 310–316.

Tassinari, W. S., Pellegrini, D. C., Sá, C. B., Reis, R. B., Ko, A. I., & Carvalho, M. S. (2008). Detection and modelling of case clusters for urban leptospirosis. *Tropical Medicine & International Health, 13*, 503–512.

Tian, Q., Ren, F., Hu, T., Liu, J., Li, R., & Du, Q. (2016). Using an optimized Chinese address matching method to develop a geocoding service: A case study of Shenzhen, China. *ISPRS International Journal of Geo-Information, 5*, 65.

Ward, M. H., Nuckols, J. R., Giglierano, J., Bonner, M. R., Wolter, C., Airola, M., ... Hartge, P. (2005). Positional accuracy of two methods of geocoding. *Epidemiology, 16*, 542–547.

Weber, D. (2003). *Network interactive user interface using speech recognition and natural language processing.* (Google Patents).

Wey, C. L., Griesse, J., Kightlinger, L., & Wimberly, M. C. (2009). Geographic variability in geocoding success for West Nile virus cases in South Dakota. *Health & Place, 15*, 1108–1114.

Yang, D.-H., Bilaver, L. M., Hayes, O., & Goerge, R. (2004). Improving geocoding practices: Evaluation of geocoding tools. *Journal of Medical Systems, 28*, 361–370.

Yildirim, V., Yomralioglu, T., Nisanci, R., & Inan, H. (2014). Turkish street addressing system and geocoding challenges. *Proceedings of the Institution of Civil Engineers: Municipal Engineer, 167*, 99–107.

Zandbergen, P. A. (2008). A comparison of address point, parcel and street geocoding techniques. *Computers, Environment and Urban Systems, 32*, 214–232.